Fiscal Year:	FY 2017	**Task Last Updated:**	FY 02/12/2018
PI Name:	Voorhees, Peter W. Ph.D.		
Project Title:	Coarsening of Dendritic Solid-Liquid Mixtures (CSLM): The Low Volume Fraction Limit (NNX16AR13G)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2010 Materials Science NNH10ZTT001N
Start Date:	09/01/2016	**End Date:**	08/31/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX16AR13G		
Performance Goal No.:			
Performance Goal Text:			
	NOTE: Continuation of "Coarsening of Dendritic Solid-Liquid Mixtures: The Low Volume Fraction Limit (NNX14AB73G)" with the same Principal Investigator, Dr. Peter Voorhees. Dendrites form during solidification into an undercooled melt. These dendrites possess secondary and sometimes even ternary arms. While the tip radius and tip velocity of the dendrite are set by the growth conditions, the sidebranches behind the tip undergo a coarsening process under nearly isothermal conditions. This coarsening process sets the final arm thicknesses and distances between dendrite arms in the solidified structure -- almost independent of the length-scale given by the dendrite tip. Unfortunately, the morphological evolution of these secondary dendrite arms remains poorly understood. Since there is a close relationship between the size scale of the coarsened dendritic structure and the mechanical properties of the material, a greater understanding of this process will have important practical implications.		

Task Description:	The reduced gravitational accelerations found during processing on the International Space Station (ISS) can allow for unprecedented insights into coarsening processes of solid-liquid mixtures in which the volume fraction of solid dendrites is small. In a microgravity environment it will be possible to follow the evolution of the morphology of the mixtures in the diffusive limit where the concentration field is set by the curvatures of the solid-liquid interfaces. It will also be possible to determine the evolution of the topology of the solid-liquid mixtures. For example, when secondary dendrite arms detach from the primary dendrite stem they will not sediment, and it will thus be possible to determine the rate at which this detachment process occurs and its spatial location. The topology and morphology of these dendritic mixtures will be determined using three-dimensional reconstructions and insights into the mechanisms of coarsening will be provided by phase field simulations that employ these reconstructions as initial conditions. These studies will provide insights into the dynamics of morphological evolution of these systems and will point the way to the formulation of models of this technologically important coarsening process.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Dendritic solid-liquid mixtures are ideal topologically complex systems with which to study coarsening. The microstructure is on a size scale that is amenable to experimental analysis using three-dimensional reconstructions and the thermophysical properties of many alloy systems are reasonably well known. Moreover, the coarsening of dendritic solid-liquid mixtures has major technological implications since it occurs in virtually all casting processes. In this case, a dendritic structure forms quickly and a so-called "mushy zone" is created. This mushy zone can exist often for long periods of time during which virtually every dendrite undergoes coarsening. The length-scales of the dendrites in the mushy zone, and thus those present after complete solidification, are determined by the coarsening process. These length scales set the microsegregation pattern and are intimately related to the mechanical properties of the ingot. Understanding the dynamics of coarsening will enable a close connection to be made between solidification conditions and the properties of the material.
Task Progress:	The process of fragmentation of dendrite arms during coarsening remains poorly understood. We performed isothermal coarsening experiments of dendritic solid-liquid mixtures using Pb-Sn alloys aboard the International Space Station (ISS), since arms that fission from the stem do not sediment and thus can be detected. The morphology of the structure and the number of fragments (fissioned arms) were determined using three-dimensional reconstructions. The evolution of the microstructure, change in length scale, interfacial shape distributions, number and distribution of fragments as well as the connectivity of the structures (handles) across coarsening time were measured. The experiments lead to a number of significant conclusions. First, there is a linear relation between surface area per unit volume and coarsening time t1/3. Comparing this to a 43% solid volume fraction Pb-Sn sample coarsened under1g conditions, we find that the relation is very similar, highlighting the absence of the effects of convection on the ground-based experiments. Additionally, the interfacial shape distributions of the two studies also evolve in a very similar fashion suggesting as well a very low amount of convection in the ground based experiments. Second, dendrite fragments that result from the breaking up of the dendritic structure could be found in each of the reconstructed samples, providing direct evidence of capillary induced fissioning of secondary (and tertiary) arms. Third, when scaled by the surface area per volume, we find that the number of fragments is nearly invariant with respect to coarsening time. This finding is surprising in view of the significant change in the dendritic morphology and its non-self-similarity during coarsening that is observed in the experiments. Thus it is possible to predict the number of fragments during coarsening by a measurement of the surface area per volume. Fourth, the connectivity of the structures as measured by the number of handles per volume is also found to be largely time invariant when scaled by the cube of the surface area per volume. The number of scaled handles is greater than the number of fragments implying that coalescence is more prevalent than fragmentation during isothermal coarsening. These findings confirm that while the morphology of structures may change substantially during coarsening, the structure may remain topologically invariant. It should be noted that due to the experimental setup, there is a relatively low amount of fragmentation, since only curvature driven fragmentation was present. Since capillary-driven fragmentation will always be present, other mechanisms for fragmentation mentioned above would be expected to increase the number of fragments above that seen in our experiments. We have used the experimental results as an initial condition in a phase field simulation of coarsening. The phase field analysis allowed us to demonstrate a mechanism of coalescence and the formation of handles during coarsening. While previous research has focused extensively on the evolution of the neck radius during the pinching process, our analysis showed the importance of the evolution of the bulb radius during the coarsening process. An ongoing pinching process may be halted as a consequence of the evolution of the radius of the bulb: we observed that the oval shaped bulb first became wider before becoming narrower and circular, possibly leading to a coalescence with an adjacent arm rather than a pinching. Our analyses identified key factors in the prediction of fragmentation events. The coarsening mechanisms of fragmentation, retraction, and coalescence have been identified for quite a while, but predicting which one will take place in a particular coarsening process has been particularly difficult. This was stressed recently by Beckermann and co-workers. The analysis of the coarsening dynamics in the original microstructure and one eroded by one voxel show that small changes in initial conditions can lead to a very different coarsening behavior. Using the experimentally measured microstructure in a phase field simulation provides new insights into the fragmentation process. The analysis confirmed the importance of secondary arm length and spacing to predict fragmentation, but equally demonstrated the importance of incorporating the complexity of dendritic structures, especially the angle of the dendrite arms, their uneven spacing and length, and their uneven shapes and coarsening rates. Our work with the European Space Agency (ESA) CETSOL (Columnar-Equiaxed Transition in SOLidification processing) team has shown a very complicated dendritic structure during the dendritic to equiaxed transition. This data was taken using a state-of-the-art serial sectioning process. Our analysis shows that the dendrites are not easily quantified in terms of envelopes of growing equiaxed dendrites. Reference Neumann-Heyme H, Eckert K, Beckermann C. 2015. “Dendrite fragmentation in alloy solidification due to sidearm pinch-off.” Phys. Rev. E 92, 060401. (https://)

Bibliography Type:	Description: (Last Updated: 01/17/2024)
Articles in Peer-reviewed Journals	Cool T, Voorhees PW. "The evolution of dendrites during coarsening: Fragmentation and morphology." Acta Materialia. 2017 Apr 1;127:359-67. https://doi.org/10.1016/j.actamat.2017.01.029 , Apr-2017
Articles in Peer-reviewed Journals	Cool T, Voorhees PW. "Dendrite fragmentation: an experiment-driven simulation." Philos Trans A Math Phys Eng Sci. 2018 Feb 28;376(2113):20170213. Invited paper. https://doi.org/10.1098/rsta.2017.0213 ; PubMed PMID: 29311211; PubMed Central PMCID: PMC5784103 , Feb-2018
Articles in Peer-reviewed Journals	Arai N, Stan T, Macfarland S, Voorhees PW, Muyanja NS, Shahani AJ, Faber KT. "Coarsening of dendrites in solution-based freeze-cast ceramic systems." Acta Materialia. 2021 Aug 15;215:117039. https://doi.org/10.1016/j.actamat.2021.117039 , Aug-2021